

## 16.1: Discrete Random Variables:

A discrete random variable is the outcome of an experiment which has an integer, numerical value. There are several values a DRV can take, & these can be listed with their respective probabilities. A probability distribution can show this.

Hence $\sum_x P(X=x) = 1$

If each outcome has the same probability, it has a uniform distribution.

Mean value Derivation:

$x_1 | P(X=x_1)$, $x_2 | P(X=x_2)$, ...

Let $x_i$ happen $n_i$ times

$\Rightarrow \sum x = \sum x_i n_i$

$\Rightarrow \bar{x} = \frac{\sum x}{n} = \frac{\sum x_i n_i}{n}$

$\frac{\sum x_i n_i}{n} = \sum x_i \cdot \frac{n_i}{n} = \sum x_i P(X=x_i)$

Variance Derivation:

$\text{Var}(X) = E[(x-\mu)^2] = \frac{\sum (x-\mu)^2}{n}$

$= \frac{\sum x^2 - 2\mu\sum x + \sum \mu^2}{n}$

$= \frac{\sum x^2}{n} - 2\mu^2 + \mu^2 = \frac{\sum x^2}{n} - \mu^2$

$= E(x^2) - E(x)^2$

Decimals for frequencies are treated as probabilities on calculators!

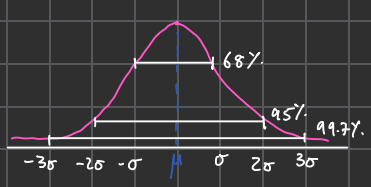
## 16.2: The Normal Distribution:

The normal distribution is a specific continuous random variable with a probability density function of:

$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

This means $f(x) > 0 \ \forall x$, $\int_{-\infty}^{\infty} f(x)\,dx = 1$, and the graph is symmetric about $\mu$.
Models continuous data, and hence can take infinite values.
For the CRV $X$:

$X \sim N(\mu, \sigma^2)$, where $\mu = E(X)$ and $\sigma^2$ = variance. For a larger variance, the graph grows wider.




To calculate probabilities, use $N_{CD}$, and hence to find:

$P(X \leq t) = \int_{-\infty}^{t} f(x)\,dx$

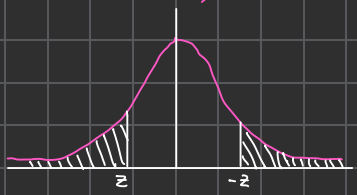
$P(X \geq t) = \int_{t}^{\infty} f(x)\,dx$

## Standardising the Normal Distribution:

A standard curve has $\mu=0$, and $\sigma=1$, and hence: $Z \sim N(0,1)$

To convert a general result:

$$z=\frac{x-\mu}{\sigma} \quad | \quad X \sim N(\mu, \sigma^2)$$




$\phi(z)$ can then be calculated, which is: $\frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z} e^{-\frac{1}{2}x^2} dx$

$\Rightarrow \phi(-z) = 1-\phi(z)$

Given a certain probability, this can be used to find the value of which would produce this probability.
This can be used on $P(X \leq x)$, $P(x_1 \leq X \leq x_2)$, $P(X \geq x)$

To find unknown parameters, ALWAYS convert into the standard normal distribution first, and then compare (to things such as $\phi(z)$).

## 16.2: The Normal Approximation to the Binomial:

A normal distribution can approximate a binomial distribution given:

$n \geq 20$ & $p \approx 0.5$ / $np \geq 10$, $n(1-p) \geq 10$

Hence: $X \sim N(np, np(1-p))$

Since Normal & Binomial are continuous & discrete respectively, continuity corrections must be added to the Normal

$P(X \geq k) \Rightarrow P(X \geq k-0.5)$ $\quad P(X>k) \Rightarrow P(X > k+0.5)$

$P(X \leq k) \Rightarrow P(X \leq k+0.5)$ $\quad P(X<k) \Rightarrow P(X < k-0.5)$

Since for Binomial

$P(X<k) = P(X \leq k-1) \Rightarrow P(X \leq k-1+0.5)$

$P(X=k) = P(k-0.5 \leq X \leq k+0.5)$